

Real and Functional Analysis Review

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Real/functional analysis for ML

- ▶ Motivation
- ▶ Function Spaces (with examples)
- ▶ Metric Spaces (with examples)
- ▶ Vector Spaces (with examples)
- ▶ Normed Vector Spaces (with examples)
- ▶ Inner Product Space (with examples)

Motivation

1. Defining and understanding matrix norms, Cauchy-Schwarz, Singular Value Decomposition, etc.
2. Reproducing Kernel Hilbert Spaces
3. Fourier Transform
4. Applications like metric learning

Function Space

It is a set of functions. Each point in the space is a function.

1. $C[a, b]$ - continuous functions on $[a, b]$
2. $L_1[a, b] - \{f : \int_a^b |f(x)| dx < \infty\}$
3. $L_2[a, b] - \{f : \int_a^b |f(x)|^2 dx < \infty\}$

Metric Space

It is a pair (χ, ρ) where χ is a space, ρ is a meric. A distance metric is any function statisfying

1. $\rho(a, b) \in \mathbb{R}_0^+$ for all $a, b \in \chi$
2. $\rho(a, b) = \rho(b, a)$
3. $\rho(a, b) = 0$ iff $a = b$
4. $\rho(a, c) \leq \rho(a, b) + \rho(b, c)$

Metric Space (examples)

1. $(\mathbb{R}, |y - x|)$
2. $(\mathbb{R}^+, |y - x|)$
3. $(\mathbb{R}^+, |\ln(y/x)|)$
4. $(L_2(-\infty, \infty), \sqrt{\int_{-\infty}^{\infty} (f_1(x) - f_2(x))^2 dx})$
5. $(\mathbb{R}^{m \times n}, \text{rank}(Y - X))$
6. (String, Levenshtein/editdistance)
7. (Graph, Shortestpathdistance)